

Partial Differential Equations

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1 Introduction

1.1 Basic Concepts

Definition 1.1. (*Partial Differential Equation*)

Let $x = (x_1, \dots, x_n)$ and let $u = u(x)$ be the unknown function. Write

$$Du = (u_{x_1}, \dots, u_{x_n}), \quad D^k u = \left(\frac{\partial^k u}{\partial x_1^{k_1} \dots \partial x_n^{k_n}} \right)_{k_1 + \dots + k_n = k}.$$

An equation

$$F(x, u, Du, D^2 u, \dots, D^N u) = f(x)$$

is a partial differential equation of order N , denoted by $Gu = f(x)$.

Definition 1.2. (*Multi-index*)

A multi-index is $\alpha = (\alpha_1, \dots, \alpha_n)$ with length $|\alpha| = \alpha_1 + \dots + \alpha_n$ and

$$\partial^\alpha = \frac{\partial^{|\alpha|}}{\partial x_1^{\alpha_1} \dots \partial x_n^{\alpha_n}}.$$

Definition 1.3. (*Homogeneous Equation*)

The equation $Gu = f(x)$ is homogeneous when $f \equiv 0$, and nonhomogeneous when $f \neq 0$.

Definition 1.4. (*Linear Equation*)

The equation is linear when $G(au + bv) = aGu + bGv$ for constants a, b . In multi-index form,

$$\sum_{|\alpha| \leq N} A_\alpha(x) \partial^\alpha u = f(x).$$

Otherwise the equation is nonlinear.

Definition 1.5. (*Semilinear, Quasilinear, and Fully Nonlinear Equations*)

The principal part is the collection of terms that contain a derivative of order N .

- The equation is semilinear when the principal part is linear, with coefficients depending only on x :

$$\sum_{|\alpha| = N} A_\alpha(x) \partial^\alpha u + A_0(x, u, Du, \dots, D^{N-1}u) = f(x).$$

- The equation is quasilinear when it is linear in the derivatives of order N , whose coefficients may depend on derivatives of lower order:

$$\sum_{|\alpha| = N} A_\alpha(x, u, Du, \dots, D^{N-1}u) \partial^\alpha u + A_0(x, u, Du, \dots, D^{N-1}u) = f(x).$$

- The equation is fully nonlinear when a derivative of order N enters nonlinearly.

1.2 Typical Equations

Definition 1.6. (*The Wave Equation*)

Let $a > 0$ and let $\Delta = \partial_{x_1 x_1} + \dots + \partial_{x_n x_n}$. The wave equation is

$$u_{tt} - a^2 \Delta u = f$$

with initial conditions

$$u(x, 0) = \varphi(x), \quad u_t(x, 0) = \psi(x).$$

Definition 1.7. (*The Heat Equation*)

Let $a > 0$. The heat equation, also called the diffusion equation, is

$$u_t - a^2 \Delta u = f$$

with initial conditions

$$u(x, 0) = \varphi(x).$$

Definition 1.8. (*Laplace's Equation*)

Laplace's equation is $\Delta u = 0$, and a solution of Laplace's equation is harmonic. The equation does not involve time, so there is no initial condition.

Definition 1.9. (*Boundary Conditions*)

Let $\Gamma = \partial\Omega$, let $\mathbf{n}$ be the outward unit normal, and let $\sigma > 0$. Dirichlet, Neumann, and Robin conditions are respectively given by

$$u|_{\Gamma} = g, \quad \left. \frac{\partial u}{\partial \mathbf{n}} \right|_{\Gamma} = g, \quad \left. \left(\frac{\partial u}{\partial \mathbf{n}} + \sigma u \right) \right|_{\Gamma} = g.$$

1.3 Second-Order Linear Equations

Definition 1.10. (*Discriminant and Type*)

Let $a_{11}^2 + a_{12}^2 + a_{22}^2 \neq 0$, and let $u = u(x, y)$ satisfy

$$a_{11}u_{xx} + 2a_{12}u_{xy} + a_{22}u_{yy} + F(x, y, u, Du) = 0.$$

The discriminant is $\Delta = a_{12}^2 - a_{11}a_{22}$, where

- $\Delta > 0$: the equation is hyperbolic;
- $\Delta = 0$: the equation is parabolic;
- $\Delta < 0$: the equation is elliptic.

Remark.

From here on the equation is linear,

$$Lu = a_{11}u_{xx} + 2a_{12}u_{xy} + a_{22}u_{yy} + b_1u_x + b_2u_y + cu = f,$$

with coefficients depending on (x, y) . When a change of independent variables $\xi = \xi(x, y)$, $\eta = \eta(x, y)$ is invertible near (x_0, y_0) , the chain rule gives

$$u_x = u_{\xi}\xi_x + u_{\eta}\eta_x, \quad u_y = u_{\xi}\xi_y + u_{\eta}\eta_y,$$

$$u_{xx} = u_{\xi\xi}\xi_x^2 + 2u_{\xi\eta}\xi_x\eta_x + u_{\eta\eta}\eta_x^2 + u_{\xi}\xi_{xx} + u_{\eta}\eta_{xx},$$

$$u_{xy} = u_{\xi\xi}\xi_x\xi_y + u_{\xi\eta}(\xi_x\eta_y + \xi_y\eta_x) + u_{\eta\eta}\eta_x\eta_y + u_{\xi}\xi_{xy} + u_{\eta}\eta_{xy},$$

$$u_{yy} = u_{\xi\xi}\xi_y^2 + 2u_{\xi\eta}\xi_y\eta_y + u_{\eta\eta}\eta_y^2 + u_{\xi}\xi_{yy} + u_{\eta}\eta_{yy}.$$

Substitution produces

$$\bar{a}_{11}u_{\xi\xi} + 2\bar{a}_{12}u_{\xi\eta} + \bar{a}_{22}u_{\eta\eta} + \bar{b}_1u_{\xi} + \bar{b}_2u_{\eta} + \bar{c}u = \bar{f},$$

with $\bar{c} = c$, $\bar{f} = f$, and

$$\bar{a}_{11} = a_{11}\xi_x^2 + 2a_{12}\xi_x\xi_y + a_{22}\xi_y^2,$$

$$\bar{a}_{12} = a_{11}\xi_x\eta_x + a_{12}(\xi_x\eta_y + \xi_y\eta_x) + a_{22}\xi_y\eta_y,$$

$$\bar{a}_{22} = a_{11}\eta_x^2 + 2a_{12}\eta_x\eta_y + a_{22}\eta_y^2,$$

$$\bar{b}_1 = a_{11}\xi_{xx} + 2a_{12}\xi_{xy} + a_{22}\xi_{yy} + b_1\xi_x + b_2\xi_y,$$

$$\bar{b}_2 = a_{11}\eta_{xx} + 2a_{12}\eta_{xy} + a_{22}\eta_{yy} + b_1\eta_x + b_2\eta_y.$$

Lemma 1.1.

If φ solves the principal equation above and $\varphi_x^2 + \varphi_y^2 \neq 0$, then

$$a_{11}(\mathrm{d}y)^2 - 2a_{12}\mathrm{d}x\mathrm{d}y + a_{22}(\mathrm{d}x)^2 = 0.$$

Proof.

Along $\varphi = C$ one has $\varphi_x \mathrm{d}x + \varphi_y \mathrm{d}y = 0$. The slope is $\mathrm{d}y/\mathrm{d}x = -\varphi_x/\varphi_y$. Then substitute $\varphi_x/\varphi_y = -\mathrm{d}y/\mathrm{d}x$ into the equation. ■

Proposition 1.1. (*Canonical Forms for Constant Coefficients*)

Suppose a_{ij} , b_j , and c are constant. A linear change of (x, y) puts the equation in one of the following forms, with constant coefficients on the right-hand side.

- Hyperbolic, first canonical form: $u_{\xi\eta} = Au_{\xi} + Bu_{\eta} + Cu + D$.
- Hyperbolic, second canonical form: $u_{rr} - u_{ss} = A^*u_r + B^*u_s + C^*u + D^*$.
- Parabolic: $u_{\eta\eta} = Au_{\xi} + Bu_{\eta} + Cu + D$.
- Elliptic: $u_{rr} + u_{ss} = Au_r + Bu_s + Cu + D$.

Example.

The equation $4u_{xx} + 5u_{xy} + u_{yy} + u_x + u_y = 2$ has $\Delta = (5/2)^2 - 4 > 0$, so it is hyperbolic. The characteristic equation $4(\mathrm{d}y)^2 - 5\mathrm{d}x\mathrm{d}y + (\mathrm{d}x)^2 = 0$ factors as $(4\mathrm{d}y - \mathrm{d}x)(\mathrm{d}y - \mathrm{d}x) = 0$, with characteristic lines $4y - x = C_1$ and $y - x = C_2$. Set $\xi = 4y - x$ and $\eta = y - x$. Then $\bar{a}_{11} = \bar{a}_{22} = 0$, $\bar{a}_{12} = -9/2$, $\bar{b}_1 = 3$, and $\bar{b}_2 = 0$, so $-9u_{\xi\eta} + 3u_{\xi} = 2$, that is

$$u_{\xi\eta} = \frac{1}{3}u_{\xi} - \frac{2}{9}.$$

Writing $w = u_{\xi}$ gives $w_{\eta} - w/3 = -2/9$. Multiplication by $e^{-\eta/3}$ produces $w = 2/3 + F(\xi)e^{\eta/3}$, and one further integration gives

$$u(x, y) = \frac{2}{3}(4y - x) + f(4y - x)e^{\frac{y-x}{3}} + g(y - x).$$

Example.

The equation $u_{xx} - 4u_{xy} + 4u_{yy} = e^y$ has $\Delta = 2^2 - 4 = 0$, so it is parabolic. The characteristic equation $(dy + 2dx)^2 = 0$ has the lines $y + 2x = C$. Set $\xi = y + 2x$ and $\eta = y$. Then $\bar{a}_{11} = \bar{a}_{12} = 0$, $\bar{a}_{22} = 4$, and the lower-order coefficients coming from ξ and η vanish, so

$$u_{\eta\eta} = \frac{1}{4}e^{\eta}.$$

Two integrations give $u = (1/4)e^{\eta} + \eta f(\xi) + g(\xi)$, hence

$$u(x, y) = \frac{1}{4}e^y + yf(y + 2x) + g(y + 2x).$$

Example.

The equation $u_{xx} + u_{xy} + u_{yy} + u_x = 0$ has $a_{12} = 1/2$ and $\Delta = (1/2)^2 - 1 < 0$, so it is elliptic. The characteristic equation $dy^2 - dx dy + dx^2 = 0$ has roots

$$\lambda_{1,2} = \frac{1}{2} \pm i\frac{\sqrt{3}}{2}.$$

Set $\xi = y - x/2$ and $\eta = -(\sqrt{3}/2)x$. Then

$$\bar{a}_{11} = \bar{a}_{22} = \frac{3}{4}, \quad \bar{a}_{12} = 0, \quad \bar{b}_1 = -\frac{1}{2}, \quad \bar{b}_2 = -\frac{\sqrt{3}}{2},$$

and the equation becomes

$$u_{\xi\xi} + u_{\eta\eta} = \frac{2}{3}u_{\xi} + \frac{2}{\sqrt{3}}u_{\eta}.$$